

LoRa E-Textile Jacket
Prototype Instruction Manual

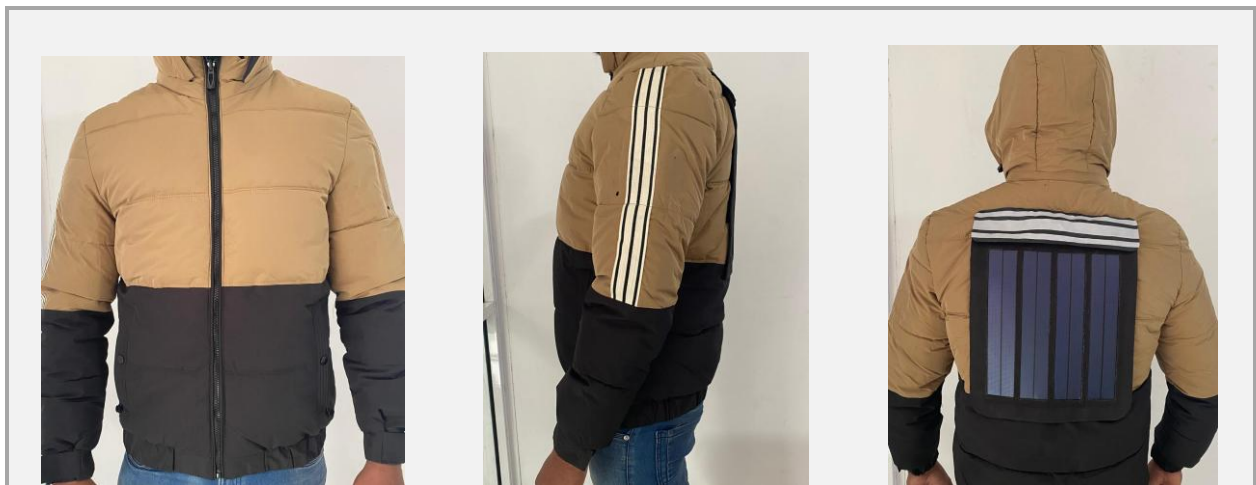
Demonstration and Evaluation Unit - Not for Sale

Product Name: Smart LoRa Jacket

Prototype ID / Build: 1.0

Manual Version: 1.0

Release Date: 2026-01-23



WARNING: Prototype demonstration unit
This jacket is a prototype intended for demonstration and evaluation only.
No safety, regulatory, or durability certifications have been obtained.
This prototype is not approved for commercial sale or consumer distribution.

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1. About This Manual

This manual describes basic setup and demonstration use of the LoRa E-Textile Jacket prototype, including module attachment, optional solar panel attachment, Android app pairing, and LoRa demonstration features.

1.1 Intended Use

This unit is provided strictly for demonstration and evaluation. It is not intended for hard use, safety-critical applications, or any commercial deployment.

1.2 Prototype Limitations

	WARNING: Not a certified consumer product No safety, regulatory, or durability certifications have been obtained for this prototype. Functionality and performance may vary across conditions and over time. Do not assume compliance with FCC/IC/CE/UKCA or other regional approvals.
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1.3 Symbols Used

This manual uses the following callout types:

	WARNING: Potential hazard / risk of injury Follow instructions to reduce risk.
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	CAUTION: Risk of damage to prototype or accessories Handle carefully to avoid damage.
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	NOTE: Helpful information Tips to improve demonstration results and usability.
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2. Critical Safety and Prototype Warnings

	WARNING: Demonstration use only Not for hard use (sports, high-motion tasks, industrial use, or harsh environments). Not intended for emergency response, safety monitoring, or medical decisions. Stop use immediately if any component appears damaged or overheats.
	CAUTION: Mechanical handling limits (prototype) Avoid excessive stretching, twisting, bending, pulling, or compressing the jacket - especially around electronics, antenna, and connector areas. Do not place heavy loads in pockets that may stress internal wiring. Do not pull the power module by its cable or connector; detach by holding the module body.
	WARNING: No washing or drying Do NOT wash or dry this jacket. This prototype has not been designed or validated for washing or drying; water exposure may permanently damage electronics and textile components.

2.1 Medical and Data Disclaimer

Sensor readings (heart rate, SpO2, skin temperature, GPS) are for demonstration only and must not be used for diagnosis or medical decisions. Accuracy may vary due to fit, motion, environment, and prototype hardware/software limitations.

3. Product Overview

The LoRa E-Textile Jacket is a winter jacket prototype that demonstrates sensor integration, detachable power, optional solar charging, Bluetooth app connectivity, and a textile-based LoRa antenna concept.

3.1 Key Features (Prototype)

- Heart rate sensor integrated at the left wrist area (wrap must be tightened).
- Detachable power module attached via magnetic snap inside the left lower pocket.
- Optional solar panel attachment on the back using a velcro frame and magnetic snap connector.
- Android mobile app pairing via Bluetooth to display heart rate, SpO2, skin temperature, and GPS location.
- Textile-based LoRa antenna for long-range transmission demonstration (for demonstration purposes only).
- Internal access flap providing external antenna clip point and an internal USB-C port for firmware update (advanced/service).

Figure - Labeled jacket overview (front and back)





3.2 Package Contents

Package contents may vary by prototype kit. Typical contents include:

- 3× Prototype jacket
- 3× Power module (without Battery)
- 3× Solar panel
- Textile antenna

4. Knowing Your Components

4.1 Heart Rate Sensor Area (Left Wrist)

The heart rate sensor is integrated at the left wrist. For best demonstration results, the wrist wrap must be snug so the sensor contacts skin.

Figure - Wrist sensor close-up



4.2 Power Module and Charging LED

The power module attaches inside the left lower pocket using a magnetic snap connector. This prototype includes one LED only: a Charging LED.



NOTE: Charging LED (single indicator)
LED ON: The module is charging (from solar panel or USB-C charger).
LED OFF: The module is not charging.

If the Charging LED does not turn on when charging is expected:

- Solar charging: confirm the solar panel is attached/aligned correctly and receiving sufficient light.
- USB-C charging: confirm the cable is fully inserted and the charger connection is stable.

Figure - Power module (LED and USB-C port)



5. Quick Start (Demonstration Setup)

1. If using the solar panel, attach the solar panel to the back of the jacket (Section 6).
2. Wear the jacket and tighten the left wrist wrap (Section 7).
3. Install the battery in the enclosure (Section 8).
4. Attach the power module inside the left lower pocket using the magnetic snap (Section 9).
5. If charging, verify the Charging LED behavior (Section 14).
6. Install the Android app and pair via Bluetooth (Section 10).

6. Attaching the Solar Panel (Optional Demonstration)

Attach the solar panel before wearing the jacket for the easiest alignment.



NOTE: Prototype charging performance
Solar charging performance may vary significantly based on sunlight conditions, panel angle, temperature, and prototype tolerances.

Figure - Solar panel attachment sequence

1. Identify the solar panel mounting area.



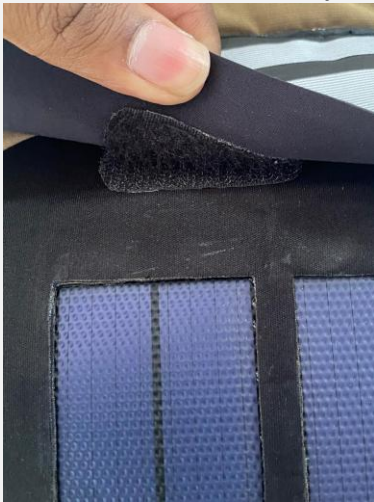
2. Align the solar panel edges and press onto the Velcro frame.



3. Connect the magnetic connector.



4. Press the two Velcro areas on the jacket to secure the solar panel frame.



5. Final view after the solar panel frame is connected to the jacket.

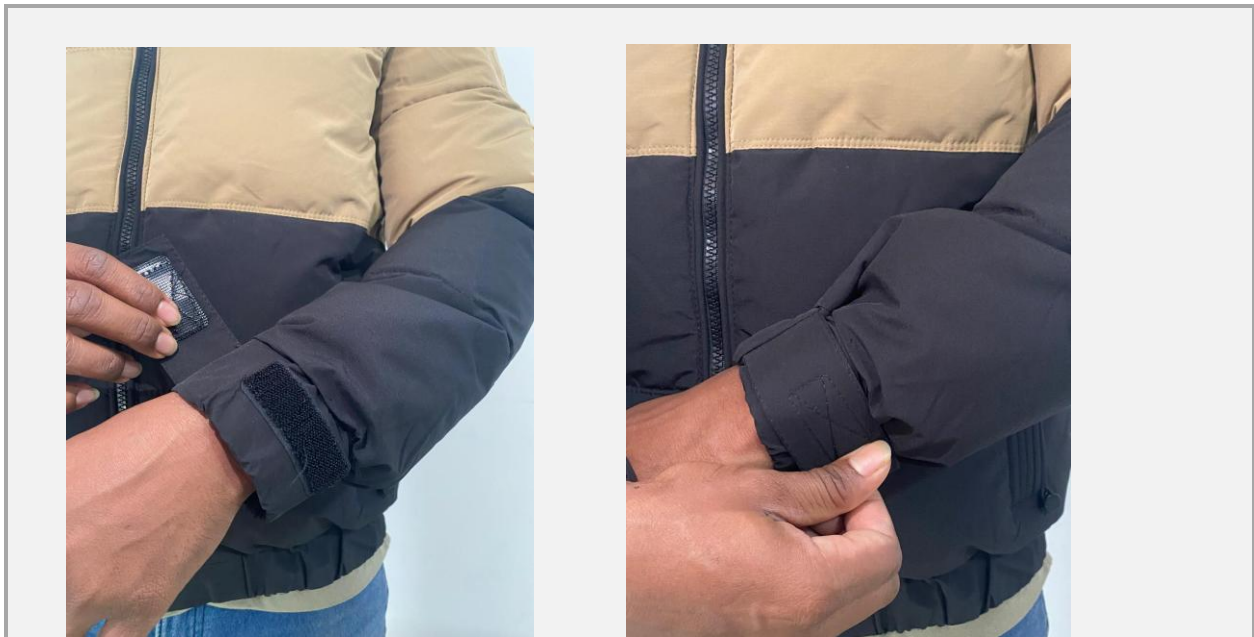


7. Wearing the Jacket and Tightening the Wrist Wrap

1. Put on the jacket normally.
2. Locate the left wrist wrap.
3. Tighten the wrist wrap until snug and comfortable (not painful).
4. Ensure the sensor area is in contact with skin (not over thick clothing).



CAUTION: Avoid overtightening
If numbness, tingling, or discomfort occurs,
loosen the wrist wrap immediately.



8. Installing the Battery in the Enclosure

Required Battery (To Be Purchased by Customer)

- Battery type: Lithium Polymer (Li-Po)
- Model: KM 103665
- Nominal voltage: 3.7 V
- Capacity: 3000 mAh
- Energy rating: 11.1 Wh

Reference example (customer may source locally):

<https://www.tokopedia.com/arenabattery/baterai-lithium-polymer-km-103665-3000mah-3-7volt-11-1wh-1731521269698102816>

Battery Installation Procedure

1. Open the Enclosure

Gently open the snap-fit enclosure by lifting along the edges.



2. Identify Battery Location

Locate the battery compartment and the 2-pin battery connector inside the enclosure.



3. Insert the Battery

Place the battery flat in its compartment without bending it.

4. Connect the Battery

Gently plug the battery connector into the matching socket on the PCB.



Connect the Battery to the PCB

The mating battery connector is on the PCB. Connect the battery-side connector to the marked connector on the PCB.

5. Secure the Battery

Arrange the wires neatly and ensure the battery does not move inside the enclosure.

6. Close the Enclosure

Press the enclosure edges together until all clips engage.

7. Power On and Verify

When the battery is connected, the module turns on automatically.

9. Attaching the Power Module (Left Lower Pocket)

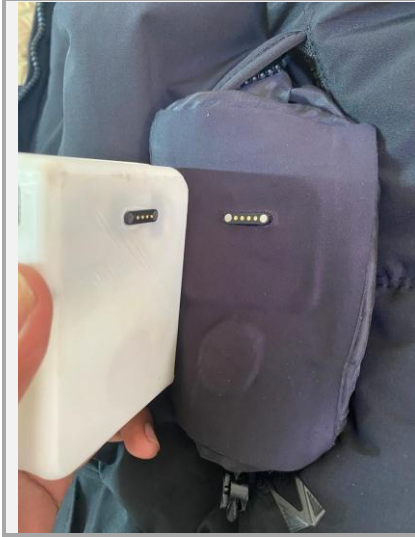
1. Open the left lower pocket.
2. Locate the jacket's magnetic connector inside the pocket.
3. Align the power module connector to the jacket connector.
4. Allow the magnetic snap to pull the module into place.
5. Confirm stable attachment before moving.



CAUTION: Prototype connector handling
Do not force the connector. If alignment does not occur easily, rotate the module and try again.
Detach gently by holding the module body-do not pull on cables or textile wiring.

Figure - Power module attachment steps





10. Mobile App Setup (Android Only - Prototype)

This prototype currently supports Android only.

10.1 Requirements

App name: Smart Jacket

Minimum Android version: Android 10 or later

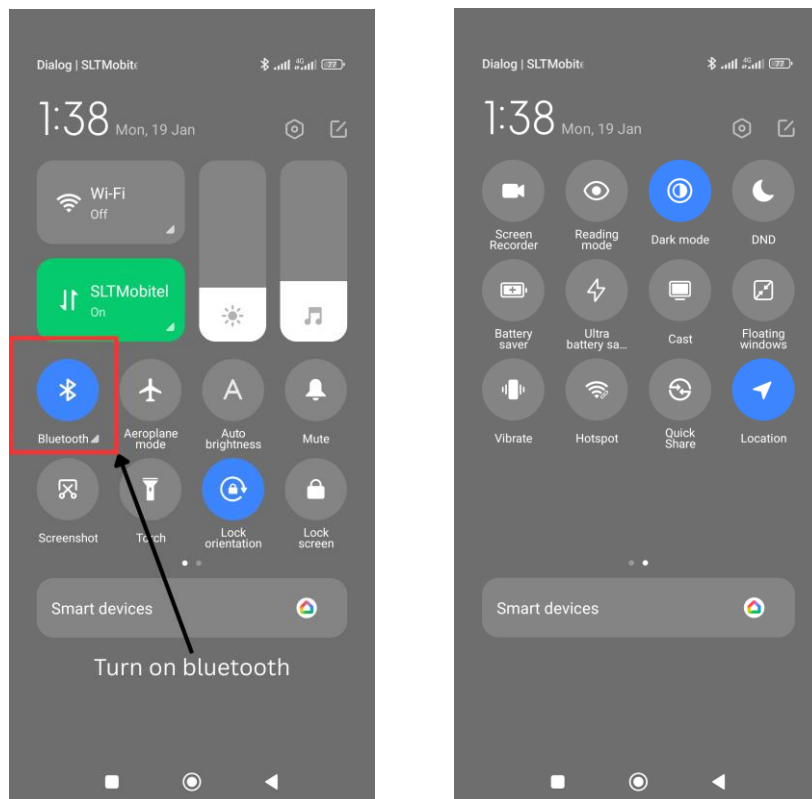
Permissions required: Bluetooth and Location

APK download link (Android):

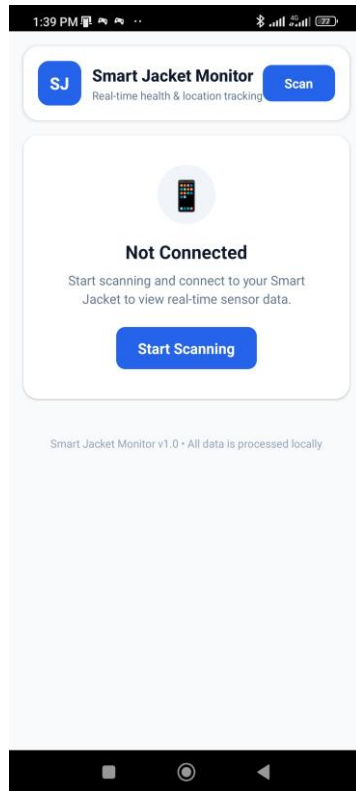
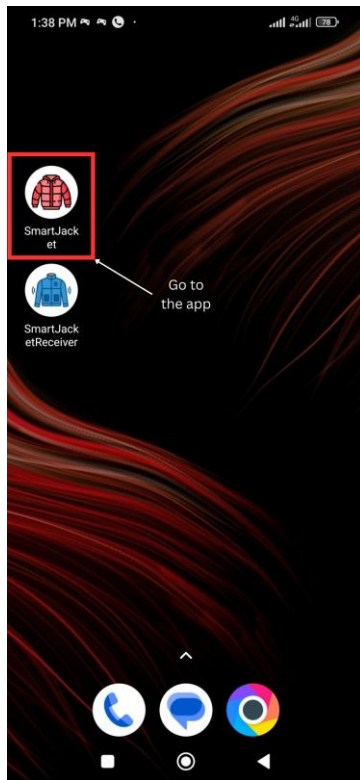
https://1drv.ms/f/c/0379e4bfab6ad848/IgAAV_UwMaBZSr_wr1is_0YCAQ1Pu0weu9SXA4Y1ZGwulNc?e=hSHFFI (Password: LoRa Jacket 2026)

10.2 Pairing Instructions

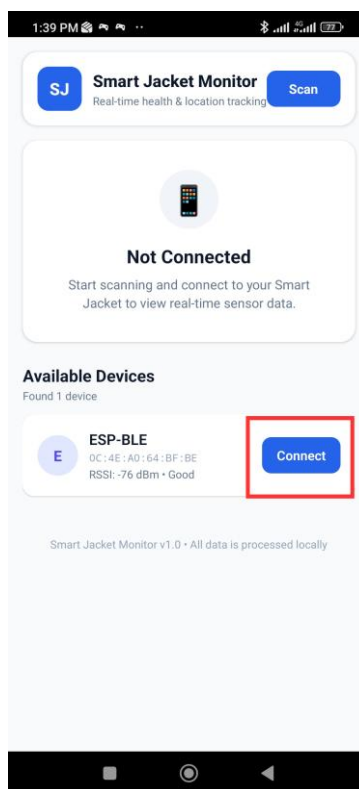
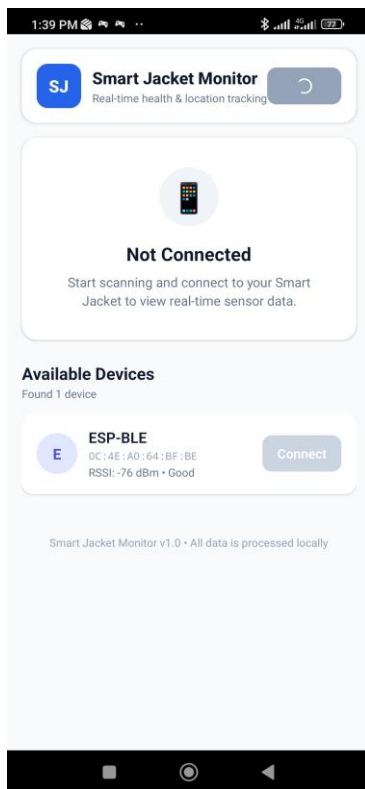
1. Ensure the power module is attached.
2. Turn on Bluetooth on your phone.



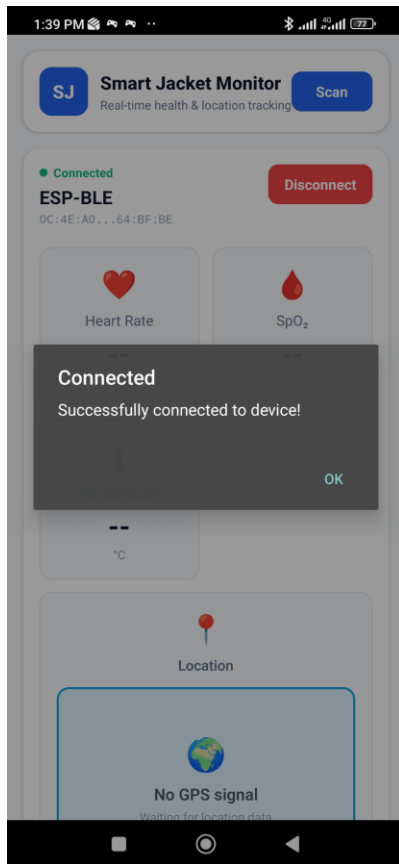
3. Open the app.



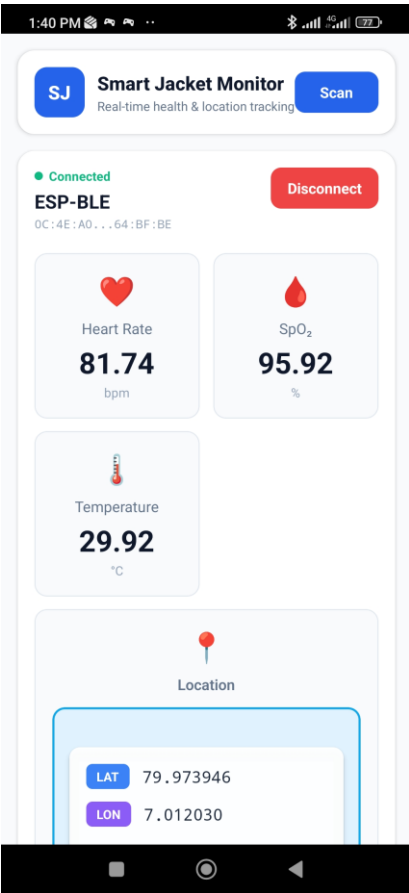
4. When prompted "Connect", tap Connect.



5. Confirm that the app shows a successful connection.



6. Verify that the dashboard shows Heart Rate, SpO₂, Skin Temperature, and GPS location. (Initial GPS lock may take 5-10 minutes.)



NOTE: Prototype connection and data
Connection stability and data accuracy may vary
due to prototype firmware and hardware
limitations.

11. LoRa Demonstration Feature (Textile Antenna)

This prototype includes a textile-based LoRa antenna intended to demonstrate the feasibility of LoRa communication with textile integration.

	WARNING: LoRa performance is not guaranteed (prototype) The textile antenna is not fully optimized and is provided for demonstration only. Antenna performance may change if the textile is stretched, folded, compressed, or if its dimensions change. No guarantee is provided for LoRa range, accuracy, or consistency in this prototype.
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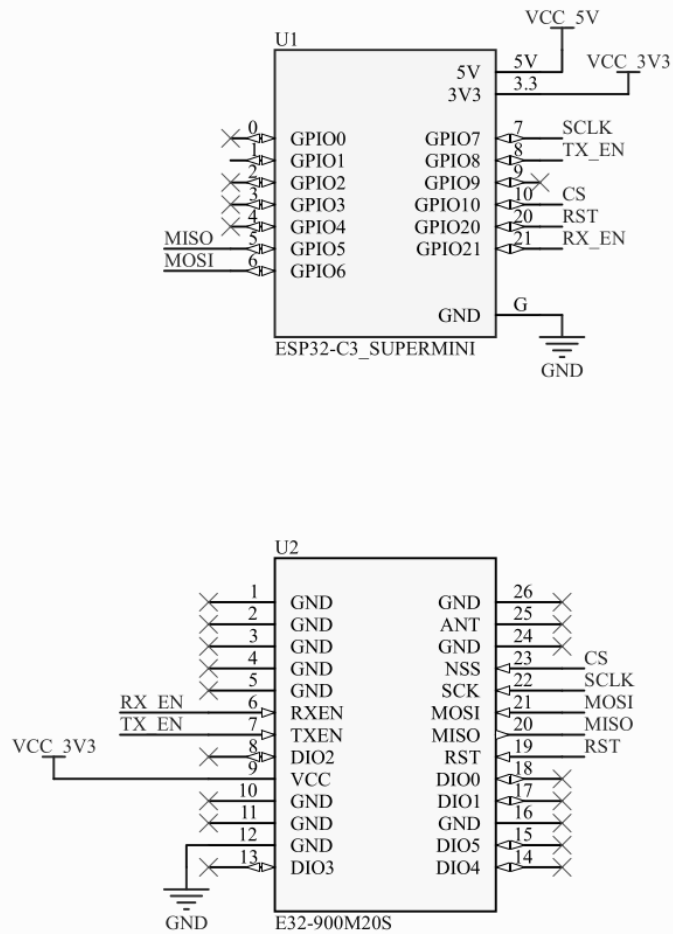
11.1 Basic LoRa Setup (Receiver Example)

LoRa mode: EBYTE E32-900M20S

Intended frequency band / region: 868 / 915 MHz

Receiver requirement: EBYTE E32-900M20S (or compatible SX1276 / RFM95-based LoRa module)

Receiver module connectivity:



Instructions:

1. Build a receiver using an ESP32-C3 module (other ESP32 modules can also be used; pin connections may vary) and an EBYTE E32-900M20S module (or another 915 MHz LoRa module such as SX1276 / RFM95; pin connections may vary by module).
2. Connect the ESP32-C3 module to the computer via USB.
3. Upload the receiver firmware to the ESP32 using Arduino IDE or VSCode + ESP-IDF.
4. Open the ESP32 serial port using a serial terminal application (for example, PuTTY) with the settings below.
 - Baud rate: 115200 bps
 - Data bits: 8
 - Stop bits: 1
 - Parity: None
 - Flow control: None

When the jacket is powered on properly, receiver data should appear in the terminal. If data is received successfully, you will see a message similar to the sample output below.

Length: 18

RSSI= -46

SNR= -11

Message (18 bytes): 05 FF 6A 00 D6 4D C4 04 22 0B A8 02 E2 03 00 02 E2 03

----- RX DATA PACKET -----

Latitude: 7.012101

Longitude: 79.973846

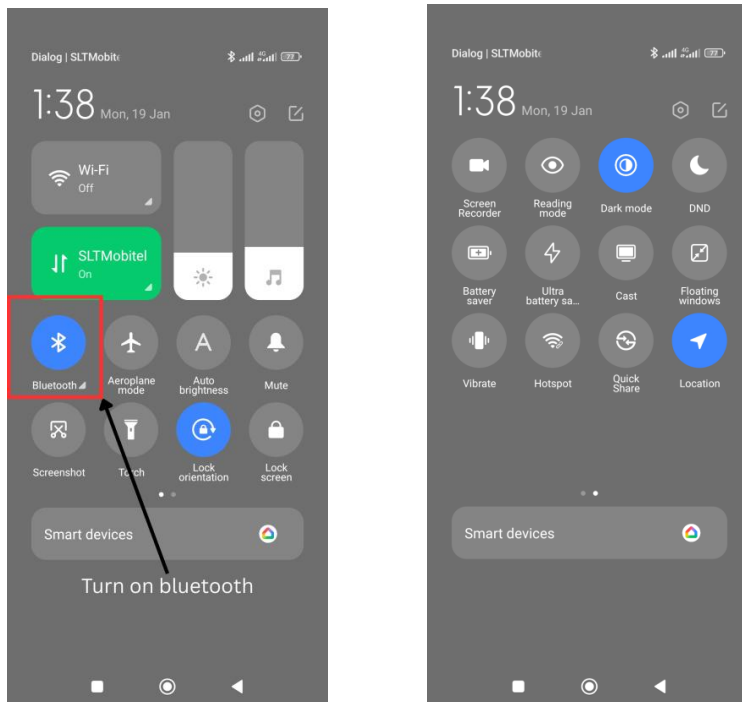
Temperature: 28.50 C

Heart Rate: 68.0 bpm

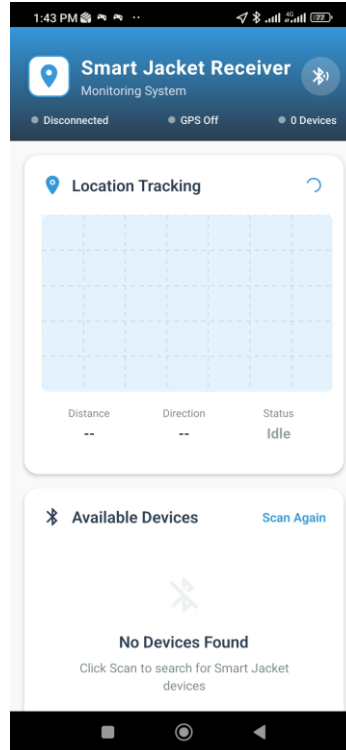
SpO2: 99.4 %

11.2 LoRa Receiver App

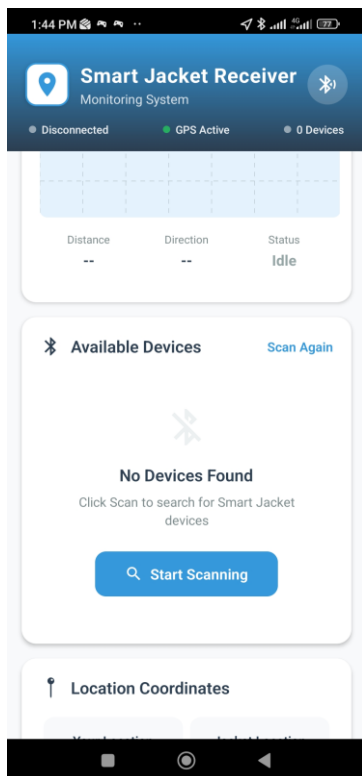
1. Turn on Bluetooth on your phone.



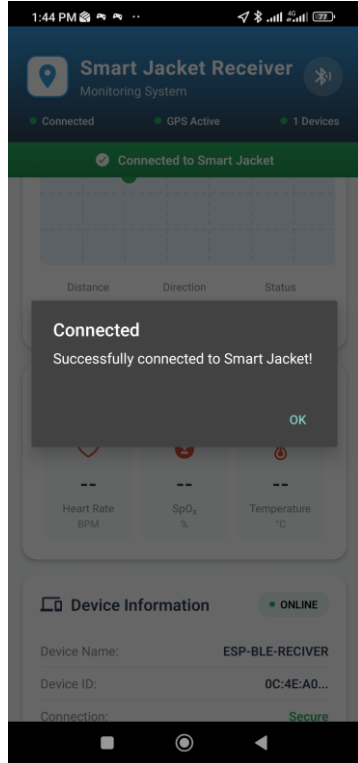
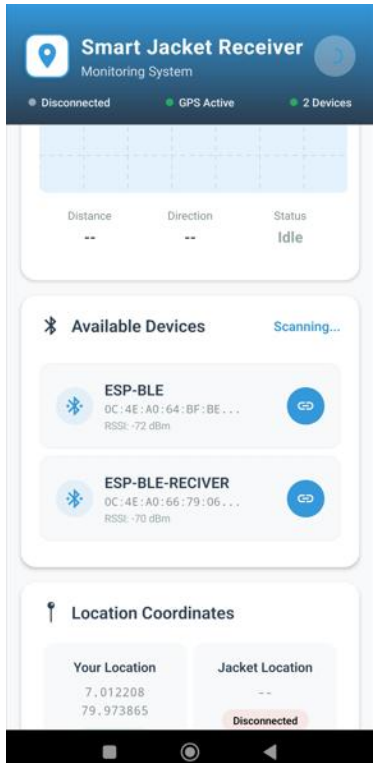
2. Open the LoRa Receiver app.



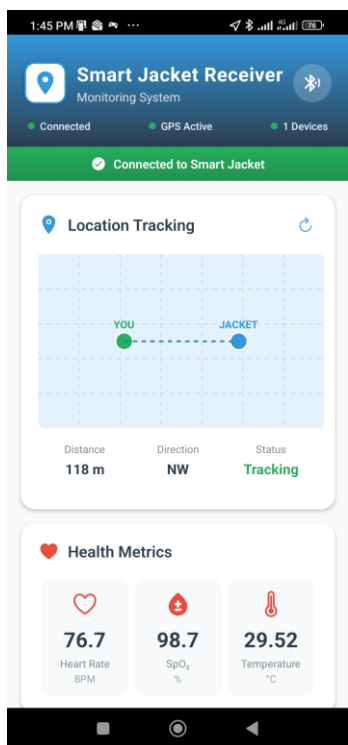
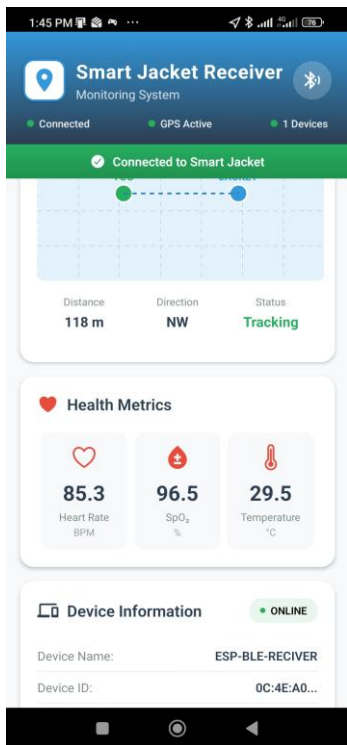
3. Tap Start Scanning.

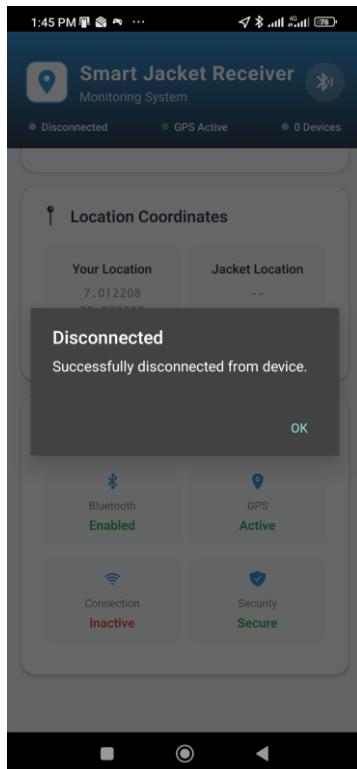
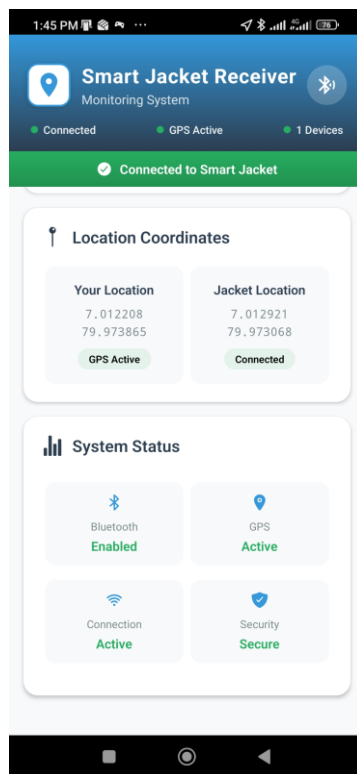
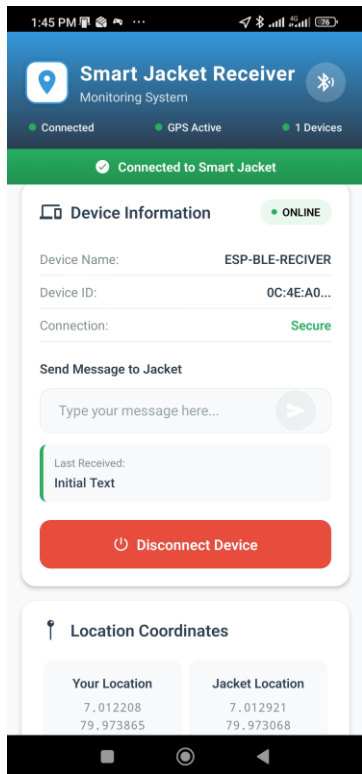


5. 4. Connect to BLE-Receiver.



5. Verify that the dashboard shows Heart Rate, SpO₂, Skin Temperature, and GPS location.





12. Internal Flap Access (Advanced / Prototype)

A small internal flap provides access to the textile antenna location, an external antenna clip point, and an internal USB-C port for firmware update.

12.1 Opening and Closing the Flap

Use the illustrated sequence below to access the internal flap. After use, close the flap fully to protect internal components.

Figure - Internal flap location and inside view

1. Reach the electronics area by inserting your hand into the left sleeve.



2. Turn the first section of the inner sleeve area outward.



3. Turn the remaining inner side area of the left arm outward.



4. Locate the internal flap inside the jacket.



5. Open the flap by lifting the fabric cover.



6. Identify the marked area.



7. Lift the marked area using your fingers.



8. Under the marked area, you can see the USB-C port.



9. After use, close the flap fully to protect internal components.

12.2 Connecting an External LoRa Antenna (Optional)

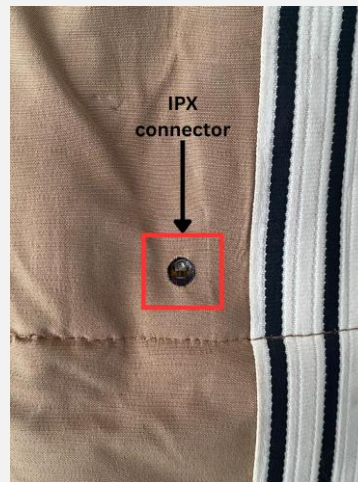


CAUTION: Prototype connector zone
Avoid repeated connect/disconnect cycles unless necessary.
Use only antennas compatible with the prototype's intended band and connector type.

Connector type/spec: IPEX connector

Figure - External antenna connector close-up

1. Identify the antenna IPX connector point.



2. Connect the antenna to the connector point.



3. Place the antenna in the pocket on the arm.



13. Firmware Update via Internal USB-C (Advanced / Service Only)



WARNING: Advanced feature
Firmware update is intended for trained personnel or as instructed by the developer. Incorrect updates may disable the prototype.

Update procedure (service reference):

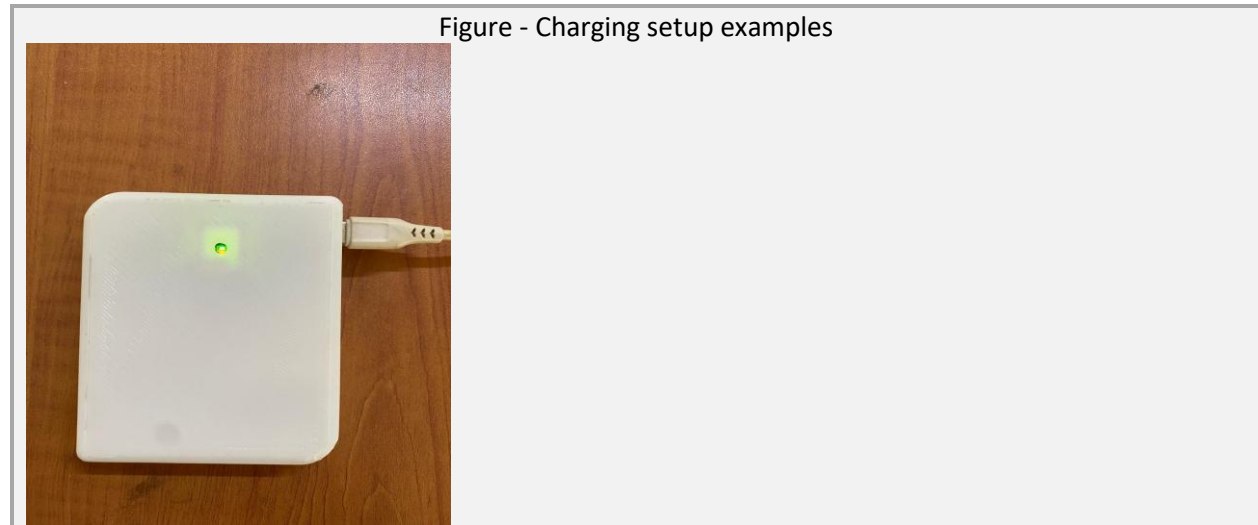
1. Open internal flap and connect the internal USB-C cable.
2. Launch the firmware update tool.
3. Select the correct device/port.
4. Load the firmware file.
5. Start the update and wait for the completion message.
6. Disconnect the cable and close the flap.

14. Charging the Power Module (Prototype)

The power module can be charged using either the solar panel (when attached) or the USB-C charging port on the module.

USB-C charging input rating: 5 V

Typical charging time (USB-C): 2-3 hours (depends on the charger)



15. Care, Handling, and Storage (Prototype)



WARNING: Do not wash or dry
This prototype is not suitable for washing or drying.
Keep electronics away from water, moisture, and cleaning liquids.

15.1 Handling Guidelines

- Avoid excessive stretching, twisting, or pulling of the jacket.
- Do not sit or place heavy objects on electronics/antenna zones.
- Open/close connectors gently; do not force magnetic snaps.
- Store flat or loosely folded; avoid sharp creases through electronics/antenna areas.

Figure - Handling do's and don'ts



16. Troubleshooting (Prototype)

Issue	Likely Cause	What to Do
Charging LED does not turn on during solar charging	Low sunlight / misalignment / snap not connected	Re-align panel; ensure snap connection; try brighter light
Charging LED does not turn on during USB-C charging	Cable not fully seated / loose connection / charger issue	Re-insert USB-C cable firmly; try another charger/cable
App won't connect	Bluetooth off / permissions not granted / prototype firmware instability	Enable Bluetooth; grant permissions; restart the app/phone
Heart rate/SpO2 unstable	Wrap too loose / motion / poor skin contact	Tighten wrap; reduce motion; ensure sensor contacts skin
LoRa range inconsistent	Textile antenna performance varies with stretch/dimensions	Avoid stretching; treat as demo-only; consider external antenna

17. Technical Specifications (Prototype)

	NOTE: Specifications subject to change Because this unit is a prototype, specifications may change without notice.
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Battery type / capacity: KM 103665 / 3000 mAh

Sensors: Heart rate and SpO2 sensor - MAX30102

Temperature sensor: TMP117

GPS module: NEO-M8N

Bluetooth version: 5.0 (LE)

LoRa module: E32-900M20S; RFM95 LoRa module - 915 MHz

Connector type for external antenna: IPX connector

18. Regulatory and Compliance (Prototype Statement)

This prototype jacket has not been tested or certified for safety, EMC/EMI, RF compliance, or consumer product standards. It is provided strictly for demonstration and evaluation use only.

19. Warranty, Liability, and Prototype Terms

This prototype is provided "as-is" for demonstration purposes only.

Appendix A - Glossary

BLE: Bluetooth Low Energy

LoRa: Long Range radio modulation used for low-power communications

LoRaWAN: A networking protocol typically used with LoRa radios

SpO₂: Peripheral capillary oxygen saturation (blood oxygen estimate)

Appendix B - Figure Index

Figures are presented inline throughout the manual.